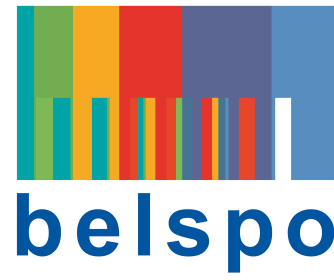




Refining the Sunspot Number Series to ISN V3: Challenges and Benefits for the Space Climate Community

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Abstract

The International Sunspot Number (ISN) remains affected by historical scale jumps such as the *Waldmeier break*, even after two major recalibrations. FARSuN (Findability and Accessibility of historical Raw Sunspot Numbers) tackles this by rebuilding the record from the ground up: collecting hand-written logs and tables from 1607–1980, digitizing them, and applying consistent metadata and validation. Early 2025 milestones include integrating the full 1610–1918 *Mittheilungen* corpus, digitizing more than 95% of the 2000+ Zürich observation tables, extracting 338 drawings (1056 entries) from the C. H. Adams notebooks, and launching quality-control pipelines for the Gruithuisen manuscripts. The reconstructed ISN V3 will deliver traceable uncertainties, hemispheric indices, and open FAIR-compliant access, strengthening space climate studies, solar cycle prediction, and coupled Sun–Earth research.

Why Reconstruction?

The ISN inherits observer-dependent scale changes (for example the 1947 "Waldmeier jump") that cannot be removed by tweaking coefficients alone. FARSuN pioneers a full reconstruction that replays the entire observing chain.

- **Recalibration** only rescales an existing index; legacy gaps, missing metadata, and hidden assumptions stay baked in.
- **Reconstruction** retrieves the primary artefacts (drawings, tabular logs, weather notes) and recomputes the index with transparent uncertainty tracking.
- Direct access to the raw sources lets us audit outliers, model observer response curves, and issue reproducible releases such as ISN V3.

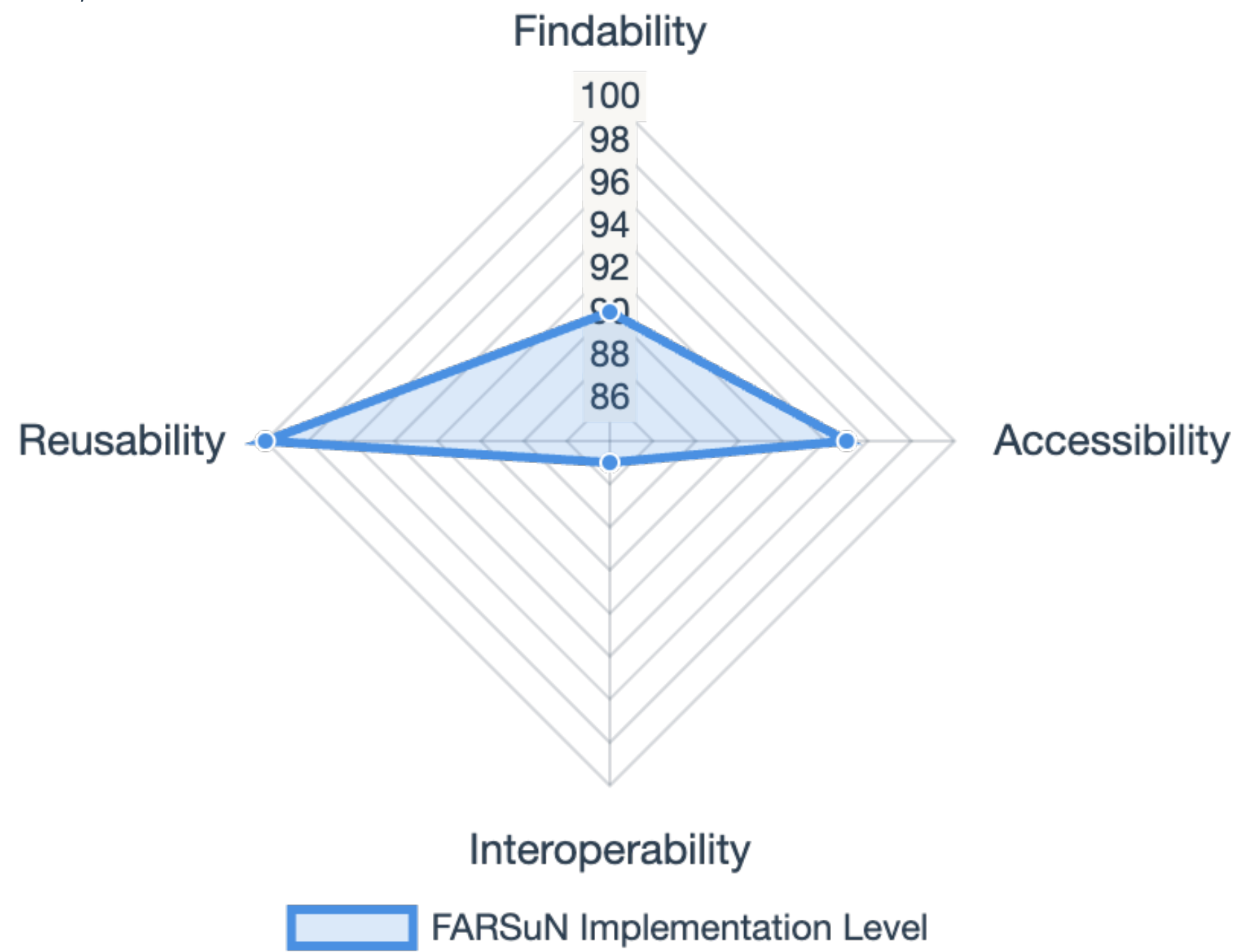
Four-Axis Methodology

1. **Gather Sources**: track down dispersed observing logs (Wolf's *Mittheilungen*, Zürich tables, Gruithuisen manuscripts, Adams drawings) and capture provenance metadata at ingest.
2. **Process Data**: combine handwriting/OCR engines (Transkribus models, custom scripts) with manual double-keying for fragile entries.
3. **Validate & Standardize**: reconcile aliases, observer IDs, and calendars; model group/spot counting behaviour to derive calibration curves.
4. **Disseminate**: publish machine-readable products, APIs, and VO-ready services with versioned uncertainty bundles for each release.

FAIR Principles

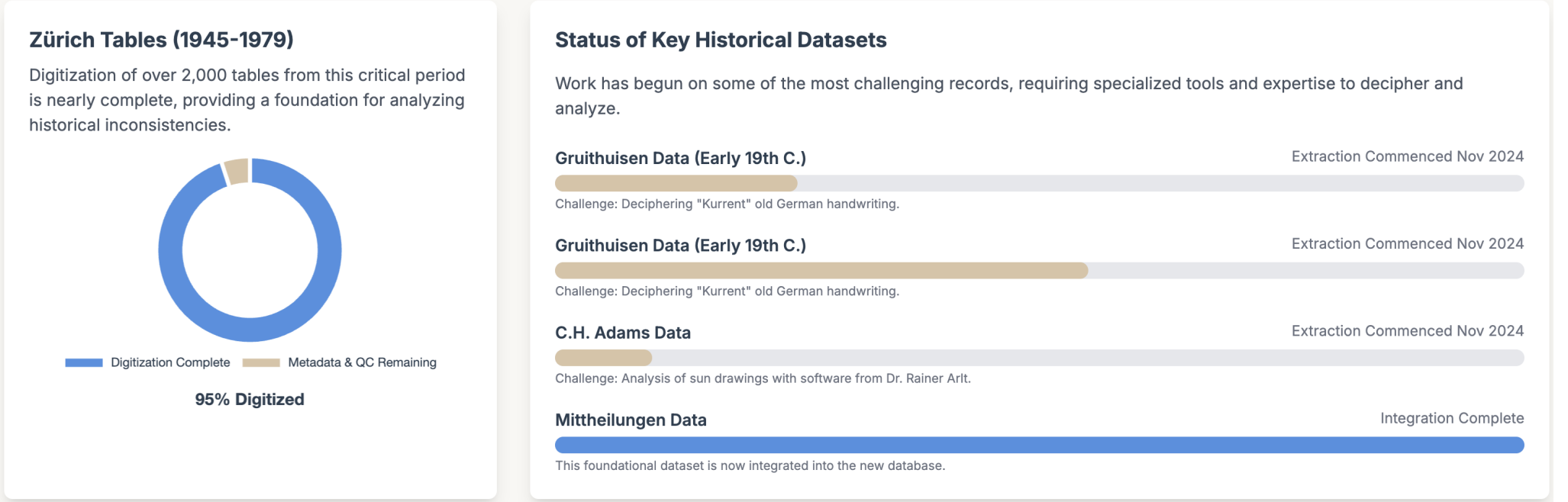
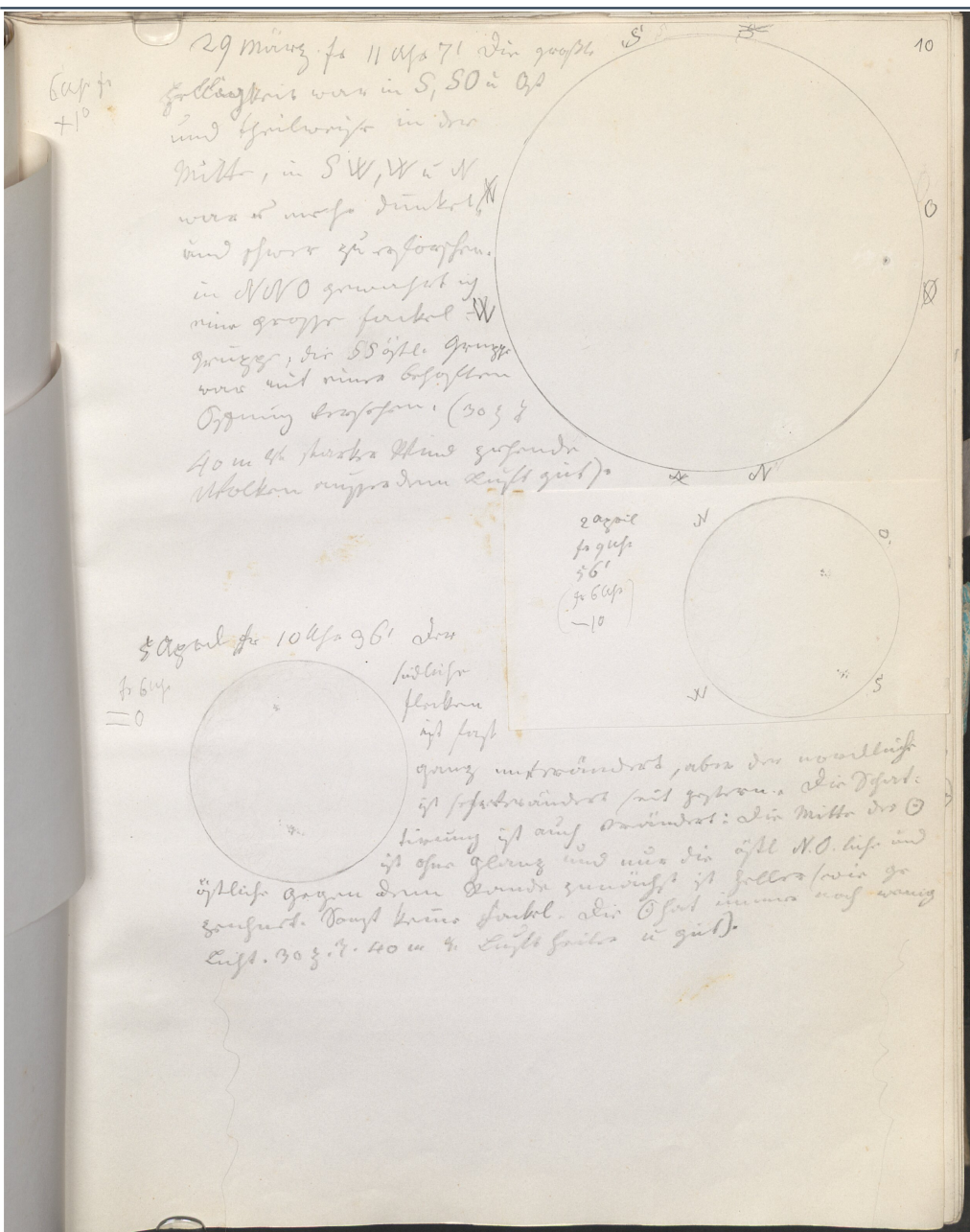
Every dataset and release roadmap is benchmarked against FAIR targets that we monitor with the dashboard shown below.

- **Findable** – persistent DOIs, rich observation-level metadata, and navigable catalogues.
- **Accessible** – open formats (CSV, NetCDF, VO tables) served via HTTPS and scripted download recipes.
- **Interoperable** – shared vocabularies, observer IDs, and VOEvent-ready semantics.
- **Reusable** – versioned releases with uncertainty models, provenance trails, and detailed caveats.



Key Datasets

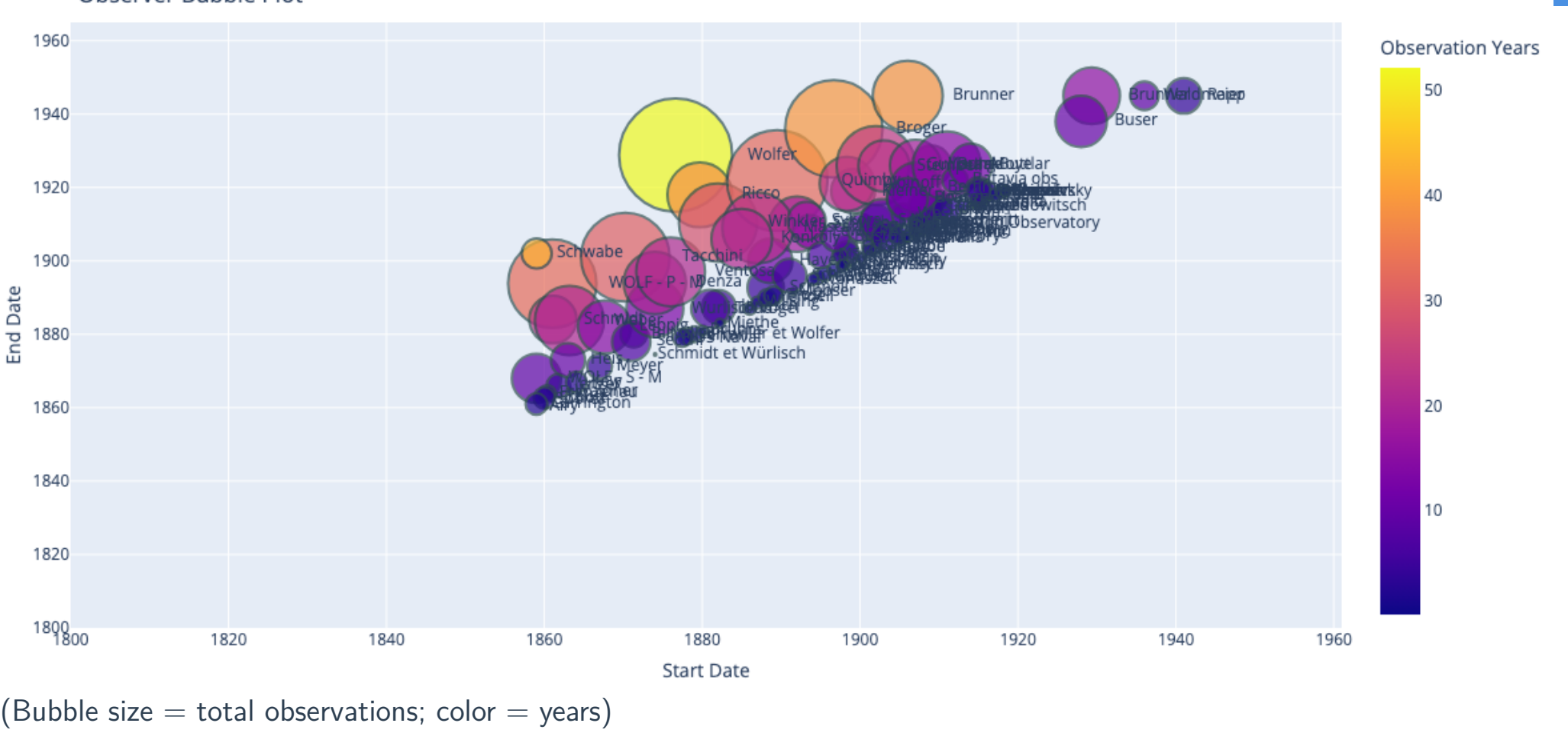
Source	Period	Current Focus
Zürich Tables	1945–1979	95% of 2000+ tables digitized; metadata encoding for late-1970s observers underway.
<i>Mittheilungen</i>	1610–1918	Fully integrated; observer statistics produced from 400+ years of entries.
Gruithuisen Manuscripts	1817–1849	25% transcription via Transkribus; file renaming algorithm + dual QC for group/spot counts.
C. H. Adams Drawings	1819–1823	338 sketches extracted (1056 entries); spotless days cross-checked against Wolf series.
Augustin Stark Ledgers	1813–1835	Normalizing tabular layouts; linking weather annotations to group counts.
Rost Journals	18th–19th c.	Building metadata vocabularies prior to digitization ingest.



Progress Highlights (Early 2025)

- Citizen-science ready interface blueprint finished; aligns with planned public transcription portal.
- Zürich production line blends Transkribus-assisted extraction with manual review to hit 2025 completion.
- Gruithuisen workflow pairs Kurrent script specialists with validation scripts to flag missing sun symbols.
- Adams drawings compared against *Mittheilungen* daily records to benchmark spotless-day agreement.

Observer Timeline



Methods

- Dual-mode transcription: tailored Transkribus handwriting models plus manual keying for complex tables.
- Data hygiene: observer alias reconciliation, calendar conversions, and spot/group recount audits.
- Uncertainty modelling: propagate observer calibration curves and note spotless-day confidence flags.
- Automation: reproducible Python pipelines with checkpoints for citizen-science ingestion and review.

From Raw to ISN V3

1. Curate daily counts with provenance-rich metadata (observer, instrument, weather context).
2. Solve observer response functions to align group/spot scales across centuries.
3. Aggregate to monthly, yearly, and hemispheric indices with propagated uncertainties.
4. Benchmark against ISN V2 and satellite-era proxies to validate drift corrections.

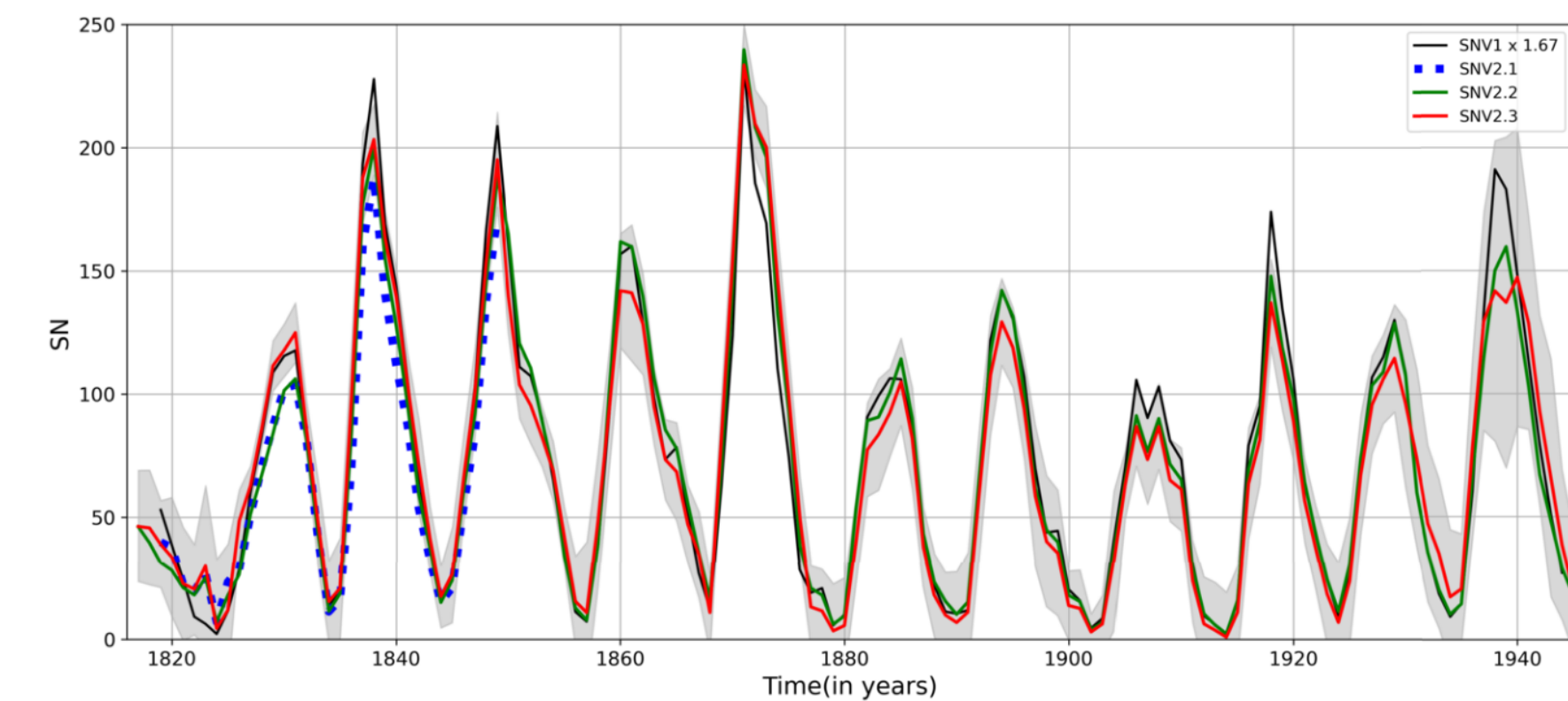


Figure 15 Annually averaged SN series. The grey-shaded area represents the 1σ uncertainty of SNV2.3. We note that SNV1 has been multiplied by 1.67 to bring it to the same scale as the other series.

Challenges

- Deciphering *Kurrent* manuscripts and annotating non-solar notes without losing context.
- Harmonising heterogeneous formats (drawings, tabular ledgers, weather marginalia).
- Quantifying observer bias and spotless-day reporting habits over centuries.
- Maintaining end-to-end provenance from scanned folio to released data product.

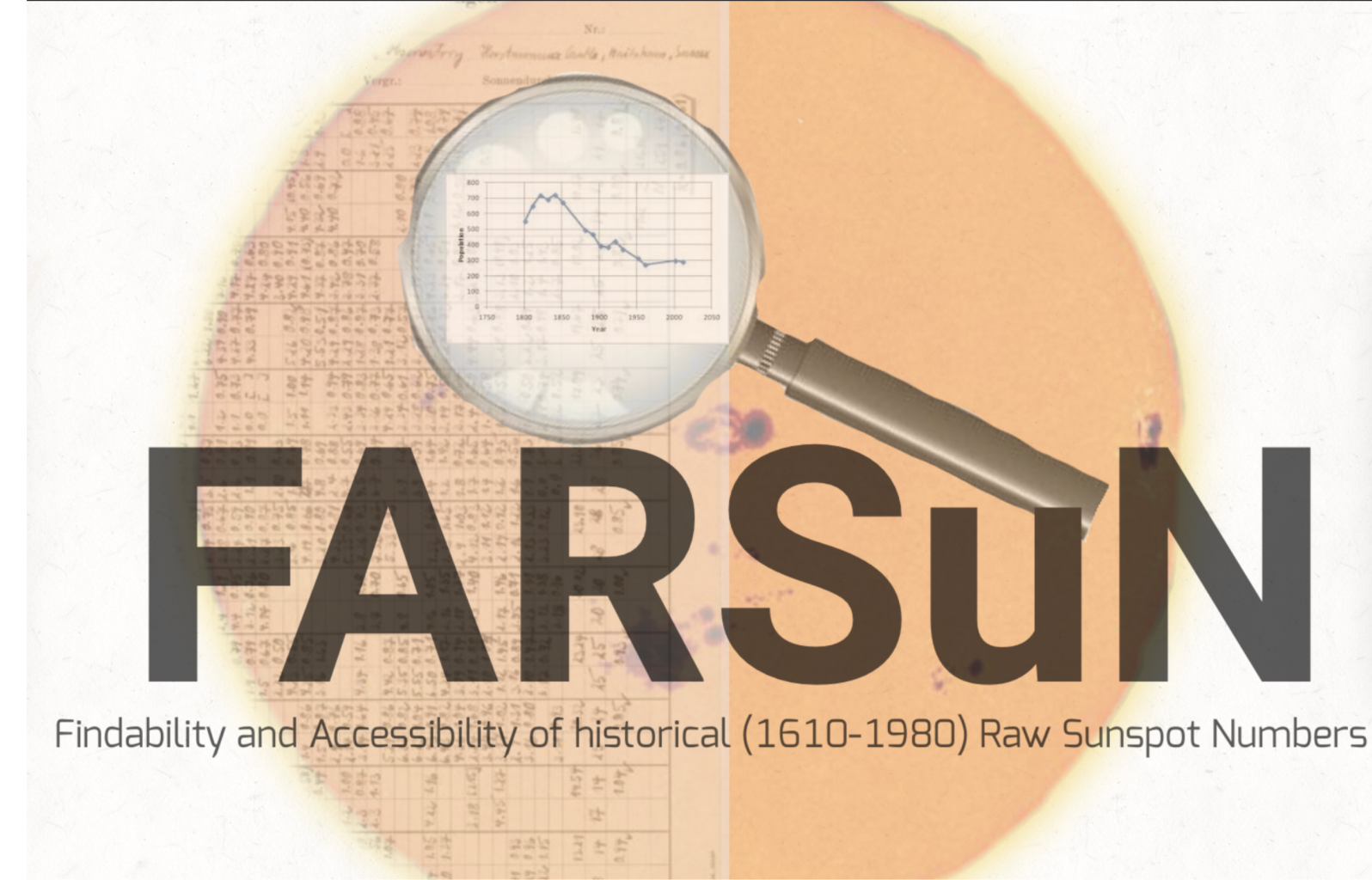
Impact

- Citizen science engagement accelerates transcription and STEM outreach.
- Improved solar cycle prediction skill via unbiased long-term baseline.
- Reference climatology for operational space weather services.
- Higher-fidelity forcing inputs for Earth system and climate models.

Outlook

- Launch citizen-science transcription portal with tailored training sets.
- Publish ISN V3 release notes with uncertainty budget and API endpoints.
- Expand collaborations (Nagoya, UCLouvain, ROB) for cross-validation and joint publications.

Findability and Accessibility of historical (1610-1980) Raw Sunspot Numbers



References